Proceedings of the 3rd International Conference on Inventive Computing and Informatics (ICICI-2025)
IEEE Xplore Part Number: CFP25L34-ART; ISBN: 979-8-3315-3830-9

OpenLearnHub: An AI-Driven Online Platform for Equitable Access to Educational Resources with Integrated NLP and Speech Recognition for Enhanced Learning Assistance

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2025 3rd International Conference on Inventive Computing and Informatics (ICICI) | 979-8-3315-3830-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICICI65870.2025.11069696

***Abstract*—Access to quality educational resources is often hindered by financial constraints and health-related challenges. OpenLearnHub is an innovative online platform designed to provide free educational materials, categorized by school levels, intermediate courses, and engineering subjects. Users can access resources through manual navigation or voice-based search, making learning more convenient and inclusive. The platform features an intelligent chatbot powered by Natural Language Processing (NLP) with Spacy, which processes voice queries to retrieve relevant textbooks, PDFs, documents, and lecture materials from a MongoDB database. Additionally, OpenLearnHub includes a Doubt Clarification Module, where users can upload doubts in text or image format, and other users can contribute by providing responses. This fosters a collaborative learning environment, enabling peer-to-peer knowledge sharing. The platform enhances engagement through personalized recommendations, interactive discussions, and AI-driven assistance, ensuring a seamless and efficient learning experience. Its user-friendly interface allows students to explore resources effortlessly, while community contributions expand the knowledge base for future learners. By leveraging artificial intelligence, cloud-based storage, and collaborative learning, OpenLearnHub removes barriers to education and makes learning more accessible, interactive, and efficient for all students.**

***Index Terms*—Voice-Based Search, Natural Language Processing (NLP), Chatbot, MongoDB Database, Study Materials, Doubt Clarification Module**

I. Introduction

Access to educational resources plays a crucial role in students' academic success, yet many, especially those from rural or underdeveloped regions, struggle to obtain the necessary materials. Small and medium-sized public secondary schools often face significant challenges in acquiring adequate teaching and learning resources, further widening the educational gap [1]. Traditional learning methods—such as relying on physical textbooks, classroom instruction, and manual library searches—can be time-consuming and inefficient. Recognizing these challenges, the proposed system provides a user-friendly online platform that allows students to access educational materials categorized by school levels, intermediate courses, and engineering subjects. By integrating advanced technologies, the platform ensures that learning resources are accessible to all students, regardless of their geographical or socioeconomic background.

The platform leverages Natural Language Processing (NLP) and speech recognition, enabling students to search for textbooks and study materials using voice commands. This feature significantly enhances accessibility, especially for students with disabilities or those with limited access to typing devices. Additionally, a Doubt Clarification Module allows users to submit doubts along with images for better clarity, ensuring a more intuitive and interactive learning experience. Other users can respond with explanations, fostering a collaborative learning environment where students engage in peer-to-peer knowledge sharing.

The system's backend is developed using Flask, a lightweight Python web framework, and MongoDB, a NoSQL database, ensuring efficient storage and retrieval of educational content. The frontend, built with HTML, CSS, and JavaScript, provides an intuitive and engaging user experience. Spacy, an advanced NLP library, processes natural language queries to deliver accurate search results and personalized recommendations. The platform also integrates cloud-based storage for seamless multi-device access while maintaining security and scalability. With multilingual support, OpenLearnHub aims to remove

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barriers to education, making learning more inclusive, accessible, and efficient for students worldwide.

II. LITERATURE REVIEW

Tregubov, V et al. [2] developed an AI-powered voice recognition tool using Google Assistant to support academic English learning during COVID-19. The system, called "Academic Phrase Bank", includes two specialized voice actions to generate and practice academic expressions. Their methodology involved designing interactive speech-based modules for grammar correction, vocabulary building, and phrase memorization. This approach aims to reduce educational inequality by making remote learning more accessible and personalized. Jariwala, A et al. [3] introduces an AI-driven digital teaching platform for visually impaired students using open-source APIs like NLTK. It features an intelligent conversational agent for personalized tutoring and a mathematical expression parser for practicing problems.

Niu, S et al. [4] examined user experiences with an AI-aided educational platform aimed at supporting personalized and socially interactive learning. Their methodology involved qualitative analysis of feedback from students, teachers, and a principal using the SLP platform. The system integrated adaptive learning tools, communication features, and content customization to enhance engagement. Findings emphasized the need for individualized learning pathways and the role of learning theories in AI-based education.

Roshanaei, M et al. [5] investigates the use of AI-driven personalized learning to advance educational equity by identifying trends in student learning and modifying course materials accordingly. It examines a range of AI applications, such as predictive analytics, intelligent tutoring, and adaptive learning systems, while also addressing issues like algorithmic bias and data privacy. In order to reduce educational gaps, the study offers practical suggestions and outlines methods for maximizing AI-based interventions. Through the use of adaptive learning models, the study seeks to develop more successful and inclusive learning environments for a range of student demographics.

Sajja, R et al. [6] introduces the AI-Enabled Intelligent Assistant (AIIA), a tool that uses natural language processing (NLP) and artificial intelligence (AI) to improve individualized and flexible learning in higher education. In order to give automatic help and feedback, it examines the system architecture and describes how AIIA easily interfaces with Learning Management Systems (LMSs). The study also looks at how AI-enabled Virtual Teaching Assistants (VTAs) can boost learning results overall, increase student engagement, and help with real-time query resolution. AIIA seeks to develop a more effective and engaging digital learning environment by utilizing AI.

Srivastava, S et al. [7] Using Deep Learning and Computer vision techniques, they proposed a cost-effective and integrated Smart Learning Assistance(SLA) tool for Education. It comprises speech to text and sign language conversion for hearing impaired students, sign language to text conversion for speech-impaired students, and Braille to text for communicating with visually impaired students.

Kingchang, T et al. [8] developed an AI-powered chatbot platform integrated with the Line application to provide educational recommendations for higher education aspirants. Their methodology involved expert evaluation and student testing, focusing on platform functionality, usability, and decision-making accuracy based on user aptitudes. The chatbot effectively analyzed queries and guided 86.15% of users in their academic decisions. High satisfaction and efficiency levels confirmed its potential as a mobile decision-support tool for personalized educational guidance.

George, R et al. [9] The OpenWHO platform, which provides free, low-bandwidth, offline-accessible online courses to support equitable worldwide learning, is examined in this research. It uses survey and platform data to examine access discrepancies by gender, country income, and preferred language. In order to effectively reach various groups, the study emphasizes the significance of flexible, culturally relevant, and asynchronous learning.

Patil, S et al. [10] proposes an AI-driven online platform that provides text summarization, question generation, and question answering to enhance e-learning. It uses transformer models like T5, DistilBERT, and DistilBART for text summarization, question-answering, and question generation tasks based on their advantages.

Themistocleous et al. [11] Open Brain AI is a platform for AI research that analyzes written and spoken language for clinical and educational purposes by combining machine learning, natural language processing, and speech recognition. Using deep learning models, it detects language deficits and facilitates acoustic and linguistic analysis as well as speech-to-text transcription. Additionally, the platform makes multimodal data integration possible for thorough diagnosis and study.

Dwivedi, Y. K. et al. [12] present a multidisciplinary analysis of the transformative potential and challenges of Artificial Intelligence across various sectors. The paper explores AI's disruption in industries like finance, healthcare, manufacturing, and logistics through algorithmic innovations and autonomous systems. It emphasizes both opportunities and risks, outlining a research agenda spanning business, government, and science. The study underscores the societal and industrial forces shaping AI's pace and direction of development.

Scutelnicu, L et al. [13] investigated the role of AI-

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powered platforms in enhancing academic learning, analyzing their purpose and evaluation methodologies. The study employed a literature review and surveyed 610 individuals to assess adoption, effectiveness, and learner perceptions. AI was found to significantly improve personalization, administrative efficiency, and engagement. However, the research also highlighted ethical concerns and the need to balance learner-centric and system-driven approaches.

Mathew, A et al. [14] explores an AI and NLP-based Intelligent Tutoring System (ITS) for computer education in schools. A pilot chatbot was developed using an open-source NLP library and Slack-based UI to assist students learning Scratch programming. Student queries were answered through the chatbot, and a two-stage testing process evaluated its performance. Results showed accurate ontology modeling, highlighting its potential for future cloud-based solutions.

Gyorodi, C et al. [15] conducted a comparative study between non-relational and relational databases, focusing specifically on MongoDB and MySQL. The study highlights the advantages of using MongoDB, a NoSQL database, over MySQL in terms of efficiency, particularly in the context of a forum application for personal and professional development. MongoDB was selected from various non-relational databases due to specific strengths outlined in the research. The integration of MongoDB within the forum's framework was also discussed, demonstrating its practical benefits.

III. MATERIALS AND METHODS

A. Architecture

OpenLearnHub is an intelligent educational platform that integrates AI-driven voice search and doubt clarification to enhance accessibility and collaboration among learners. The architecture consists of several core components that work together to process user queries, retrieve relevant educational resources, and facilitate community-driven doubt resolution.The arcitecture shown in Figure 1

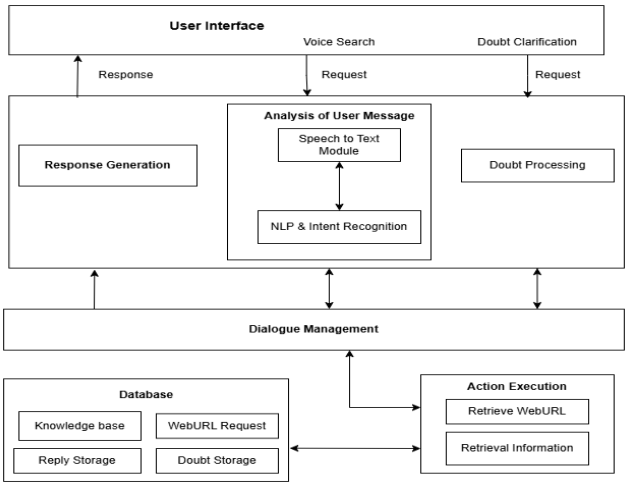


Fig. 1. Architecture

1. User Interface (UI): The User Interface (UI) is the central access point for students, enabling seamless interaction with the platform. It offers two key functionalities: Voice Search, where users can find educational resources by speaking their queries, and Doubt Clarification, which allows students to submit doubts through text or images for peer responses. The UI is designed to be intuitive, making navigation easy for users of all backgrounds. It processes user inputs, forwards them to the backend, and displays appropriate responses efficiently. Additionally, interactive elements such as voice recognition buttons and image upload options enhance accessibility and engagement.

2. Analysis of User Message: This module is responsible for interpreting user queries by leveraging Natural Language Processing (NLP) and Intent Recognition to understand their context. The Speech-to-Text Module converts spoken inputs into text, enabling seamless voice-based interactions. The NLP and Intent Recognition component, powered by Spacy, analyzes user intent to differentiate between study material requests and doubt submissions. This ensures that the platform correctly processes user inputs and routes them to the appropriate module. By accurately interpreting queries, this module enhances response accuracy and improves the overall user experience.

3. Doubt Processing and Response Generation: In the Doubt Processing module, user-submitted doubts are stored in the database, allowing other users to provide responses. The system may also suggest similar doubts or related study materials to assist users. In the Response Generation module, if the user query is a study material request, the system retrieves relevant resources from the database and presents them. These components ensure that users receive accurate study materials or community-driven solutions, fostering an interactive and collaborative learning environment.

4. Dialogue Management: The Dialogue Management module serves as the core processing unit, handling interactions between the user interface, message analysis, and backend modules. It ensures that voice search queries and doubt submissions are routed to the appropriate processing components, such as the database for storage or the action execution module for retrieving relevant study materials. This module streamlines communication within the system, ensuring smooth and efficient responses to user requests.

5. Database and Action Execution: The system utilizes a MongoDB database to store educational resources, user-submitted doubts, and responses, ensuring organized data management. The database holds a knowledge base, doubt storage, reply storage, and web URL requests for retrieving external study materials. The Action Execution module processes retrieval tasks, fetching relevant study materials

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and responses from the database. This integration enables seamless access to learning resources, making the platform efficient and user-friendly for students seeking knowledge and doubt clarification.

B. Methodology

The methodology describes how a voice-based educational support system processes user queries, retrieves study materials, and handles doubt clarification. Shown in Figure 2 The process begins when the user interacts with the system,

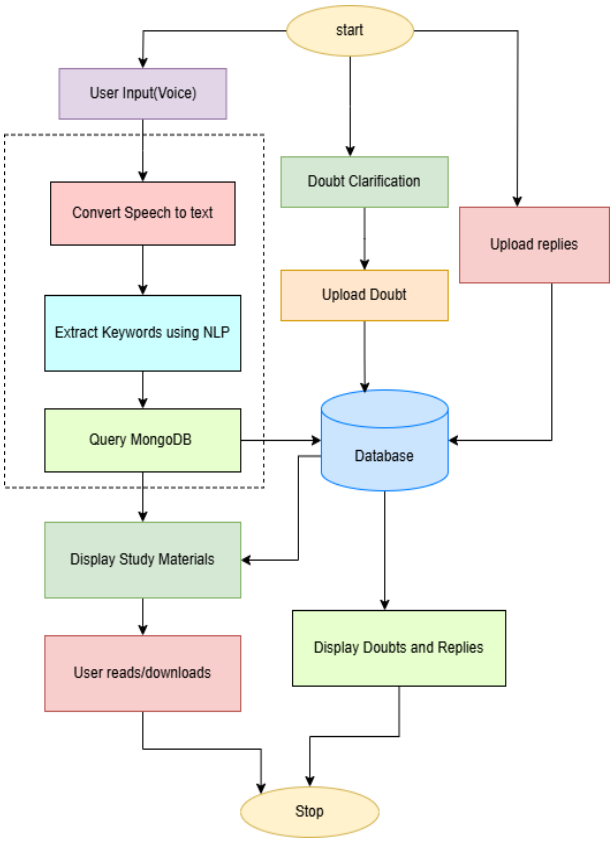


Fig. 2. Methodology

either by voice input (for study material search) or by submitting doubts.

1. User Input (Voice): When a user initiates a voice-based search, the system records the spoken query and processes it. The recorded audio is converted into text using Automatic Speech Recognition (ASR). This text is then analyzed to extract relevant keywords, which help in retrieving the appropriate study materials or resources.When a user clicks on the chatbot show in Figure 3, a new page opens where they can interact using voice input.

2.Convert Speech to Text:The system employs an Automatic Speech Recognition (ASR) module to convert spoken input into text. This conversion allows users to interact using voice commands while ensuring their queries are properly formatted for further processing. By accurately transcribing speech, the

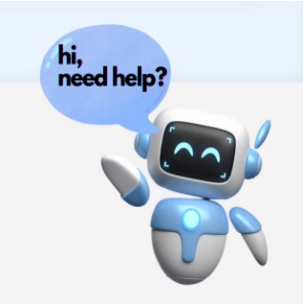


Fig. 3. ChatBot

system enhances accessibility and ease of use. The generated text is then passed to the next stage for analysis and retrieval of relevant information.

3.Extract Keywords using NLP and Query MongoDB: The system processes the converted text using NLP techniques to extract essential keywords from the user's query. These keywords are then checked against a structured CSV file that contains file paths for various study materials. The database stores these file paths, categorized by subjects and chapters. If a match is found in the CSV file, the system retrieves the corresponding study material from the database. This ensures that users receive the most relevant educational resources based on their queries.

4. Display Study Materials and User reads/downloads: Once the relevant study material is retrieved, the system displays the corresponding PDF file or resource link to the user. The user can either read the material directly within the platform or download it for offline access. This ensures a seamless learning experience by providing quick and easy access to educational resources. The interface is designed to be user-friendly, allowing students to efficiently find and utilize study materials.

5. Doubt Clarification: Doubt Clarification Page allows users to submit their doubts in text or image format for community-driven resolution. When a doubt is uploaded, it is stored in the database, making it accessible for other users to view and respond. This ensures that students receive timely and relevant answers from peers or experts. The system organizes doubts efficiently, enabling easy retrieval and categorization. By fostering collaboration, the platform enhances the learning experience through shared knowledge and interactive discussions.

6. Upload replies and Displays Doubts and Replies: Users can respond to submitted doubts by posting their answers, which are stored in the database under the relevant doubt entry. The system retrieves and displays doubts along with their corresponding replies, allowing users to view previous discussions and benefit from shared knowledge. This collaborative approach ensures that students can learn from

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each other's insights and solutions. By organizing doubts and responses systematically, the platform enhances accessibility and promotes interactive learning.

IV. RESULTS

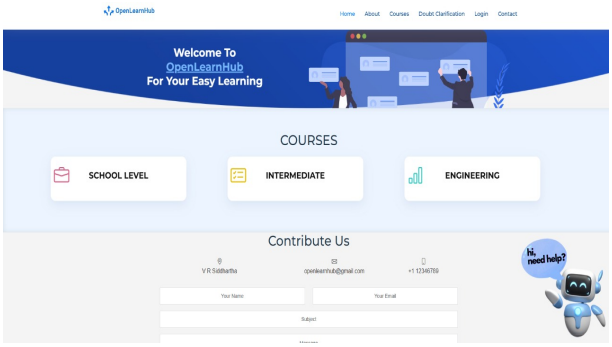


Fig. 4. Home Page

Figure 4 shows the home page of OpenLearnHub, an online learning platform. It provides course options for different education levels, including School Level, Intermediate, and Engineering. The page also features a Doubt Clarification section where users can submit their queries for peer responses. Additionally, it includes a chatbot for assistance and a section for user contributions.

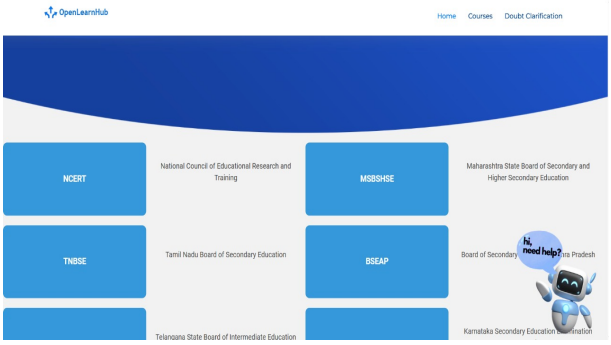


Fig. 5. Resources

Figure 5 shows the Resources page of OpenLearnHub, which provides access to educational materials from various academic boards, including NCERT, BSEAP, TSBIE, and others. Users can explore study materials based on their respective boards, enhancing their learning experience.

Figure 6 displays the class category section of OpenLearnHub, where users can select subjects based on their grade level. It provides categorized study materials for classes 6 to 10, covering subjects like Hindi, English, Social Science, and Science.

Figure 7 shows the Voice HelpBot interface in OpenLearnHub, which allows users to interact using voice commands. By clicking the "Start Listening" button, users can speak their queries, and the system will process them to provide relevant responses or study materials.

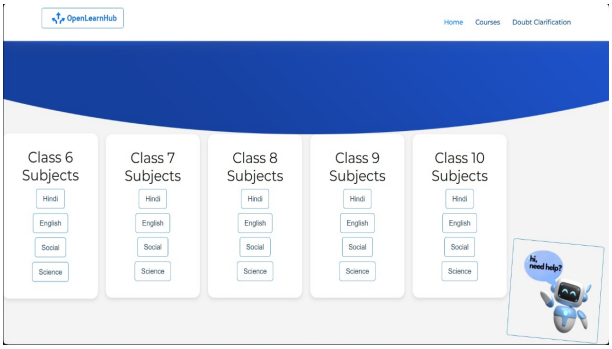


Fig. 6. Class Category

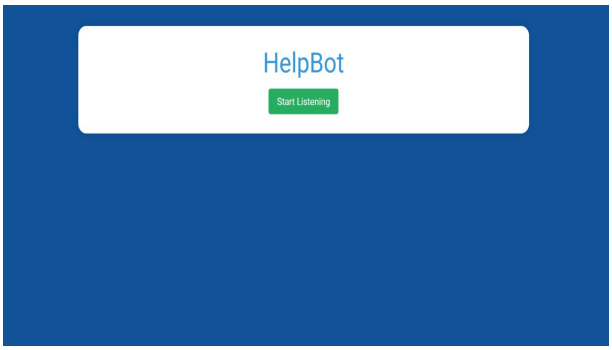


Fig. 7. Voice HelpBot

Figure 8 shows the data retrieval process using the Voice HelpBot in OpenLearnHub. After the user provides a voice query, the system processes it and displays relevant study materials, such as textbooks, based on the requested subject and standard.

Figure 9 illustrates the "Upload Doubt Clarification" interface, where users can submit their academic doubts by selecting the class and subject. They can input their doubt as text or upload an image for better clarity. The system stores these doubts for peer discussion and resolution.

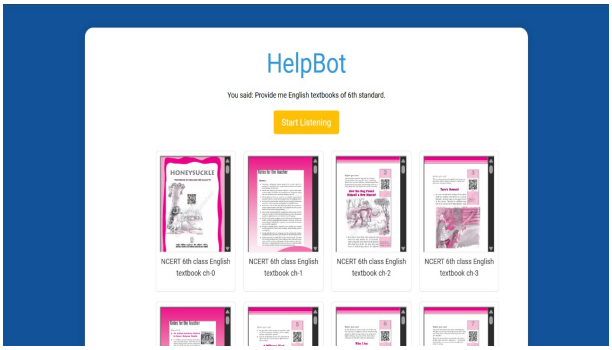


Fig. 8. Dataretrived from Voice

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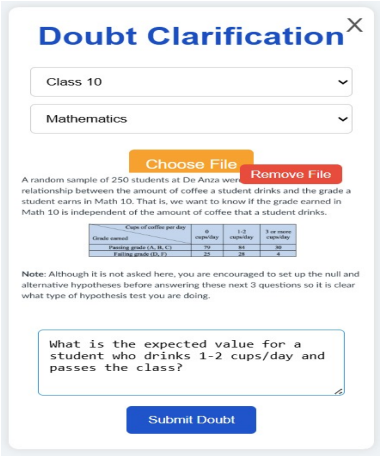


Fig. 9. Upload Doubt clarification

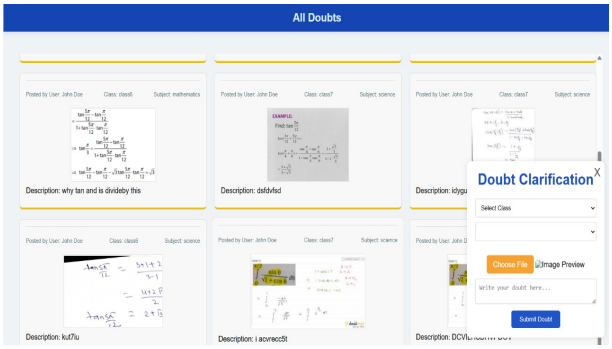


Fig. 10. Uploaded Doubts Output

Figure 10 showcases the uploaded doubts section, where users can view and access previously submitted doubts along with their descriptions. The interface also includes a "Doubt Clarification" form, allowing users to submit new doubts by selecting a class, uploading an image, and providing a description.

V. CONCLUSION AND FUTURE SCOPE

OpenLearnHub is a comprehensive online platform that uses speech recognition and natural language processing to bridge the accessibility gap in education. The platform guarantees that students from a variety of backgrounds can obtain high-quality education without facing financial or practical obstacles by offering free and organized learning resources. A collaborative doubt clarification module, chatbot support, and voice-based search are all integrated to improve user engagement and provide an engaging educational environment. The learning process is further personalized through the use of AI-driven recommendations, which increases its effectiveness and efficiency. All things considered, OpenLearnHub is an effective instrument for democratizing education and advancing inclusive learning opportunities.

In order to provide a more engaging learning environment, OpenLearnHub will be improved in the future by adding more materials, such as interactive simulations, quizzes, and video lectures. In order to customize information according to user progress and preferences, AI-driven personalized learning pathways and adaptive assessments will be implemented. In order to make educational content available to a wider audience, the site will also have bilingual support. The accuracy of voice-based search and doubt resolution will also be improved by enhancing the chatbot's natural language processing (NLP) skills with cutting-edge deep learning models. Last but not least, including blockchain-based certification and credentialing systems will aid in verifying users' learning outcomes, transforming OpenLearnHub into a comprehensive and cutting-edge learning environment.

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